

Q- List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.

Feature	Site-to-Site VPN	Remote Access VPN
Connection	Router ↔ Router	PC/User ↔ VPN Router
Purpose	Connects two entire networks/branches	Allows an individual user to access a private network remotely
VPN endpoint	Usually two routers/firewalls	One VPN server/router and a VPN client
User authentication	Usually router-to-router authentication using pre-shared keys or certificates	Usually requires username/password plus VPN group/key
Client software	Normally no special software is required on individual PCs	A VPN client is required on the user's device
Example	Head Office ↔ Branch Office	Employee laptop → Branch Office
Configuration in Packet Tracer	Both routers are configured for IPsec	Router is configured as VPN server and PC uses the built-in VPN client

Site-to-Site VPN Advantages & Disadvantages

Advantage:

Connects entire networks: All devices in both branches can communicate through the VPN.

Easy for users: Employees don't need to start a VPN connection individually because the router handles the tunnel.

Disadvantage:

More expensive infrastructure: Both locations need VPN-capable routers/firewalls.

Less suitable for individual remote users: It is designed for connecting fixed networks rather than employees working from different locations.

Remote Access VPN Advantages & Disadvantages

Advantages:

1. **Supports remote workers:** Users can access the company network from outside the office.

2. **Individual authentication:** Each user can have separate credentials, providing better control over who can access the network.

Disadvantages:

1. **Requires VPN client configuration:** Every remote device needs to be configured with the appropriate VPN settings.
2. **Depends on the user's Internet connection:** Poor or unstable Internet connectivity can affect the VPN connection and performance.